Absorbent

Your time. Your pace.



Turn lecture recordings into illustrated notes.

Ivan Mikhnovich · Egor Zybchenko · Georgy Makarov · Sergey Vazhenin

Send a recording. Get usable material.

A recording, with optional slides or board photos.

01 / SEND
Link or file

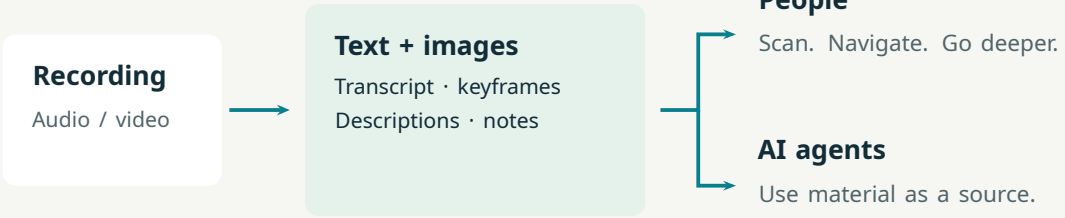
02 / PROCESS
Absorbent core

03 / READ
Illustrated notes

@absorbobot
Telegram interface

Start now. Let it process while we explain.

Useful for people. Accessible to AI agents.



From a time-based medium to material you can work with.

One recording. Two signals.

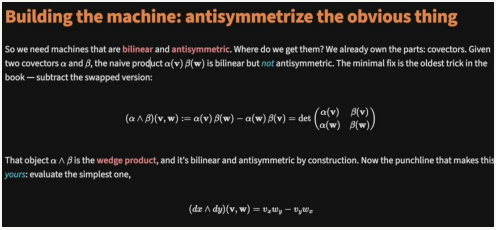
SPEECH / EXPLANATION AND LECTURE FLOW



Slides and board photos also enter the vision stage. Visual descriptions + synthesis: `deepseek-flash`.

A readable result — not just a transcript.

SOURCE VIDEO FRAME



GENERATED NOTES / EARLIER RUN

Then it reaches the **wedge product**: machines that are bilinear and antisymmetric. The minimal fix is the oldest trick in the book — antisymmetrize the obvious product:

$$(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v) = \det\begin{pmatrix} \alpha(v) & \beta(v) \\ \alpha(w) & \beta(w) \end{pmatrix}.$$

For example,

$$(dx \wedge dy)(v, w) = v_x w_y - v_y w_x .$$

Open notes excerpt / Eero Alvar · *How I Use AI to Learn Things* · 17:14

How we will measure quality.

01 / COVERAGE

Key points

Which important ideas survive?

02 / FAITHFULNESS

Correct claims

Which claims are supported and correct?

03 / VISUAL INFORMATION

Essential visuals

Which formulas and diagrams survive?

Compare with transcript-only notes. Same sources. Track time and cost.

Working today. Improving next.

WORKING TODAY

Audio / video + lecture materials

Notes: Markdown / HTML / PDF

CLI + separate Telegram bot

NEXT / EVIDENCE BEFORE COMPLEXITY

Establish an evaluation set

Compare approaches

Improve measured weaknesses

Repository · Dockerfile · `.env.example` · Documented launch

Make recordings useful. Then make learning personal.

TODAY	DIRECTION	LONG TERM
Accessible material	Source-grounded assistance	Adaptive learning

github.com/multiagent-team/absorbent

Why this pipeline?

THE MODELS

Whisper turbo

Local speech recognition.

DeepSeek `deepseek-flash`

Lecture summary, visual descriptions, final notes synthesis via API.

THE CHOICE

A fixed workflow for a known task.

No autonomous tool selection.

No training or fine-tuning.

Add agent autonomy only when it improves an evaluated outcome.

A small, auditable comparison.

01

Annotate the original recording.

Mark key points and essential visuals; do not treat STT as ground truth.

02

Compare the same source fragments.

Transcript-only notes versus the multimodal pipeline. Keep text model fixed.

03

Audit errors and trade-offs.

Coverage, unsupported / distorted claims, visual information, time and cost.

What we do not guarantee.

Perfect recognition	STT can distort speech or hallucinate. Errors can propagate into the notes.
Verified facts	Grounding instructions are not a proof. No automatic audit of every claim yet.
Cloud-free defaults	Transcripts and images reach DeepSeek. Optional cloud STT sends audio to Mistral.